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# SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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## ABSTRACT

Mixture of Experts (MoE) models now define the scaling frontier for large language models, but their mechanistic behavior cannot be read as a straightforward variant of dense networks. Prior work explains dense models through *superposition*: the use of fewer dimensions to represent more features, governed strongly by feature sparsity and feature importance. We show that this dense-model account is insufficient for MoEs. In our toy MoE setting, feature sparsity and feature importance do not produce the same discontinuous phase changes; instead, network sparsity (the ratio of active to total experts) is the decisive organizing variable. We introduce metrics for measuring superposition across experts and show that increasing network sparsity consistently increases *monosemanticity*. We further define expert specialization in terms of monosemantic feature representation rather than load balancing, and demonstrate that appropriately initialized experts organize around coherent feature combinations. These results establish network sparsity as a mechanism for more interpretable MoE representations at comparable reconstruction performance, directly challenging the assumption that interpretability and capability must trade off.

## 1 INTRODUCTION

Mixture of Experts (MoEs) have become a core architecture for state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), because they deliver computational efficiency and performance gains at scale (Shazeer et al., 2017; Fedus et al., 2022). Later work sharpened this scaling recipe through improved routing (e.g., Expert-Choice routing) and more stable training and transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical results also show that learnable routers can recover latent cluster structure in data, giving a concrete basis for expert specialization (Dikkala et al., 2023). What remains missing is not evidence that MoEs work, but a decisive mechanistic account of how they represent features differently from dense models.

Interpretability-oriented MoE work has already shown that expert behavior can be made more transparent. Yang et al. (2025b) proposes MoE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling the number of experts to enable capability editing through expert activation. These approaches demonstrate the promise of interpretable MoE design, but they primarily intervene architecturally. The unresolved mechanistic question is sharper: how do MoEs represent features, how do experts change superposition, and can specialization emerge without additional regularization? Mu & Lin (2025) identify this mechanistic interpretability problem as a central open challenge, and we address it directly.

Superposition is a central obstacle for mechanistic interpretation: models represent more features than they have dimensions. This packing strategy efficiently stores many sparse features in fewer neurons, but it also makes individual neurons polysemantic and therefore harder to interpret.

MoE architectures add the missing variable for this problem: network sparsity. Dense models activate all neurons on every input, whereas MoEs activate only a fraction of their total

parameters (Shazeer et al., 2017). This distinction changes the representational tradeoff. Dense models exploit feature sparsity by packing sparse features into shared neurons; MoEs can instead use routing to select among experts and allocate capacity to specific feature combinations.

We therefore test three concrete claims: (1) MoEs exhibit less superposition than comparable dense models, (2) MoE experts do not inherit the dense-model phase changes induced by feature importance and feature sparsity, and (3) expert specialization is better understood through feature representation than through load balancing alone.

We study these claims with simple models that extend Elhage et al. (2022)’s framework to MoEs. Our contributions are: (1) we show that MoEs replace the sharp dense-model phase changes with smoother transitions as network sparsity increases; (2) we show that MoEs consistently achieve greater monosemanticity (less superposition) than dense models with matched active and total parameter budgets, with individual experts representing features more cleanly; and (3) we introduce an interpretability-centered definition of expert specialization based on monosemantic feature representation, demonstrating that experts organize around coherent feature combinations rather than merely balancing load.

## 2 RELATED WORK

**Superposition and Feature Representations.** The Linear Representation Hypothesis frames concepts as directions in activation space (Park et al., 2024), while the number of interpretable features often exceeds the available dimensions. Elhage et al. (2022) formalized this mismatch as superposition and showed that dense models use non-orthogonal feature packing to increase capacity at the cost of polysemanticity. Superposition is theoretically efficient (Scherlis et al., 2025), but it also makes monosemantic structure difficult to recover without post-hoc methods such as Sparse Autoencoders (Bricken et al., 2023). Recent work clarifies how data correlations shape superposition (Prieto et al., 2025) and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). We extend this line of work from dense networks to MoEs and identify network sparsity—not feature sparsity alone—as the governing representational variable.

**Interpretability of MoEs.** MoEs are now a standard route to scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), yet their internal feature geometry remains less understood than that of dense models. Existing analyses emphasize macroscopic behavior, including routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). Recent interpretability-oriented approaches induce specialization through architectural constraints, including predefined concept routing (Yang et al., 2025b) and expert counts scaled to vocabulary size (Park et al., 2025). These results are valuable, but they do not explain the intrinsic geometry by which experts alter superposition. We fill that gap by measuring expert-level superposition directly, showing that MoEs are more monosemantic than dense models, and formulating specialization as feature-based occupancy plus monosemantic representation.

## 3 BACKGROUND

Mechanistic interpretability aims to reverse engineer neural networks by decomposing model representations into human-interpretable concepts, or ‘features’. We adopt the standard linear-feature framing: any hidden state  $h$  can be described as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where  $\vec{f}_i$  is the direction corresponding to feature  $i$ ,  $\alpha_i$  is the activation strength of that feature (roughly, the degree to which feature  $i$  is present in the input), and  $F$  is the set of represented features.

The *superposition hypothesis* states that models can represent far more features than dimensions, i.e.  $|F| > m$  for  $h \in \mathbb{R}^m$  (Elhage et al., 2022). Achieving this capacity requires feature

vectors in  $F$  to be packed non-orthogonally. When two features interfere,  $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$ , we say they are in *superposition*. This is a powerful capacity strategy when features are sparsely active, as in much online text data, but it directly reduces interpretability because  $\alpha_i$  then includes spurious activations unrelated to feature  $i$  being present in the input.

Monosemanticity is typically a property of individual neurons: a neuron’s activation cleanly corresponds to a single  $\alpha_i$  (i.e., features are basis-aligned). Orthogonal features that are not basis-aligned can still yield *polysemantic* neurons even when feature interference is low. Our analysis targets the interference side of this distinction. Throughout the paper, when we describe models, experts, or features as “more monosemantic,” we specifically mean that they exhibit less superposition.

## 4 DEMONSTRATING SUPERPOSITION

MoEs are often described as collections of dense models, with each expert treated as an independent dense network. Our results show that this intuition is incomplete: routing changes the feature geometry that each expert sees. We therefore compare MoEs and dense models directly in terms of superposition.

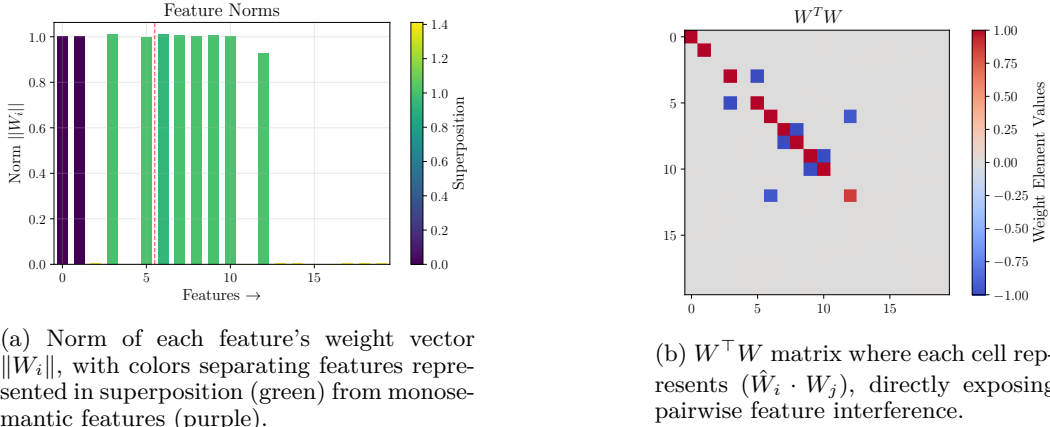


Figure 1: Feature representation and superposition in a dense model with  $n = 20$  features and  $m = 6$  hidden dimensions, with importance  $I = 0.7^i$  and uniform feature density  $(1 - S) = 0.1$ . Color reports superposition via  $\sum_j (\hat{W}_i \cdot W_j)^2$ , highlighting the dense model’s global interference structure.

### 4.1 EXPERIMENTAL SETUP

We test MoE feature geometry by asking how a MoE projects a high-dimensional vector,  $x \in \mathbb{R}^n$ , into lower-dimensional expert representations,  $h \in \mathbb{R}^m$ , and then reconstructs it. This is the MoE extension of the Elhage et al. (2022) toy-model framework.

**The input distribution.** The input vector  $x$  represents activations from an idealized disentangled model: each dimension  $x_i$  is a distinct, independent feature, giving a perfectly neuron-aligned and monosemantic source representation. We sample  $x_i \sim U(0, 1)$ , with sparsity  $S \in [0, 1)$  imposed by  $P(x_i = 0) = S$ . To encode unequal feature utility, each feature  $x_i$  receives a scalar importance  $I_i$  that weights the reconstruction loss. To isolate phase behavior for a single feature, we scale the magnitude of the last feature by  $r \in \mathbb{R}^+$ , such that  $I = (1, 1, \dots, r)$ . Together,  $I_i$  and  $r$  control feature signal strength and let us measure expert sensitivity to feature magnitude.

**Model Architecture.** The MoE has  $E$  experts, with each expert  $e$  parameterized by a weight matrix  $W^e \in \mathbb{R}^{m \times n}$  and bias  $b^e \in \mathbb{R}^n$ . A learned router  $g(x) = \text{softmax}(W^r x)$ , where  $W^r \in \mathbb{R}^{E \times n}$ , assigns each input to the top- $k$  experts. Each active expert maps the input to a lower-dimensional hidden state  $h^e = W^e x$  and produces the reconstruction

$\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ <sup>1</sup>. The model output is the weighted sum of active expert reconstructions:  $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$  where  $w_e$  are the renormalized gating weights.

We train with an L2 reconstruction loss weighted by feature importances  $I_i$ ,  $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$ . In Section 4, we add the standard auxiliary load-balancing loss (Fedus et al., 2022) to prevent expert collapse, defined as  $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$ . Here,  $N$  is the total number of experts,  $f_e$  is the fraction of batch samples routed to expert  $e$ ,  $P_e$  is the average gating probability assigned to expert  $e$  across the batch, and  $\alpha = 0.01$  sets the penalty strength. We deliberately omit auxiliary load balancing in Section 4 so that the measured superposition reflects the MoE architecture’s intrinsic bias rather than an external balancing objective.

## 4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we measure two core quantities: *representation strength* and *interference*. The expert-level norm  $\|W_i^e\|$  measures how strongly expert  $e$  represents feature  $i$ :  $\|W_i^e\| \approx 1$  indicates full representation, while zero indicates that the feature is not learned. We visualize interference between feature  $i$  and other features in expert  $e$  with the Gram matrix  $W^\top W$ , whose off-diagonal entries give pairwise interference.

Figures 1a and 2a show that the dense model and MoE represent a comparable number of features (10 vs 8) with similar norms under an equal total-parameter budget ( $m_{\text{dense}} = \sum m_{\text{experts}} = 6$ ). The crucial difference is not feature count but interference geometry. Although individual MoE experts still show local superposition (e.g., Expert 0 in Fig 2b), the global interference structure is partitioned. In the dense model (Figure 1b), any feature can interfere with any other; in the MoE, features routed to different experts have zero cross-expert interference. This establishes that MoEs allocate capacity by partitioning feature space and thereby shrink the global scope of interference.

**Expert Feature Dimensionality.** To quantify how MoEs allocate limited capacity differently from dense models, we use feature dimensionality: the “fraction of a dimension” assigned to a specific feature (Elhage et al., 2022). For feature  $i$ , we define its dimensionality in expert  $e$  by

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

$D_i^e$  ranges from zero (not learned) to one (monosemantic). We therefore define total MoE capacity as  $D = \sum_e \sum_{i=1}^n D_i^e$ .

**Efficient Packing.** Efficient packing means that total feature dimensionality matches the number of embedding dimensions, i.e.  $\sum_{i=1}^n D_i^e \approx m$  (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For MoEs, the corresponding relation is  $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$ . Empirically, both dense and MoE models satisfy this dimensionality constraint, showing that MoEs preserve packing efficiency under the same total parameter budget while changing where interference occurs.

**Features per Dimension & Network Sparsity.** Because both dense and MoE models efficiently pack features, the decisive comparison is features per dimension. This quantity directly measures superposition and reveals how the number of experts changes superposition across feature sparsities. Values above one indicate superposition because the model represents more features than available dimensions. We define MoE features per dimension by

<sup>1</sup>The ReLU at the output enforces non-negative reconstructions, matching our input distribution where  $x_i \in [0, 1]$  when nonzero. The off-diagonal terms in  $(W^e)^\top W^e$  create negative interference, and ReLU suppresses these negative components. Under sparse features, negative interference is effectively “free” because it is filtered to zero, favoring configurations with negative off-diagonal terms (e.g., antipodal pairs).

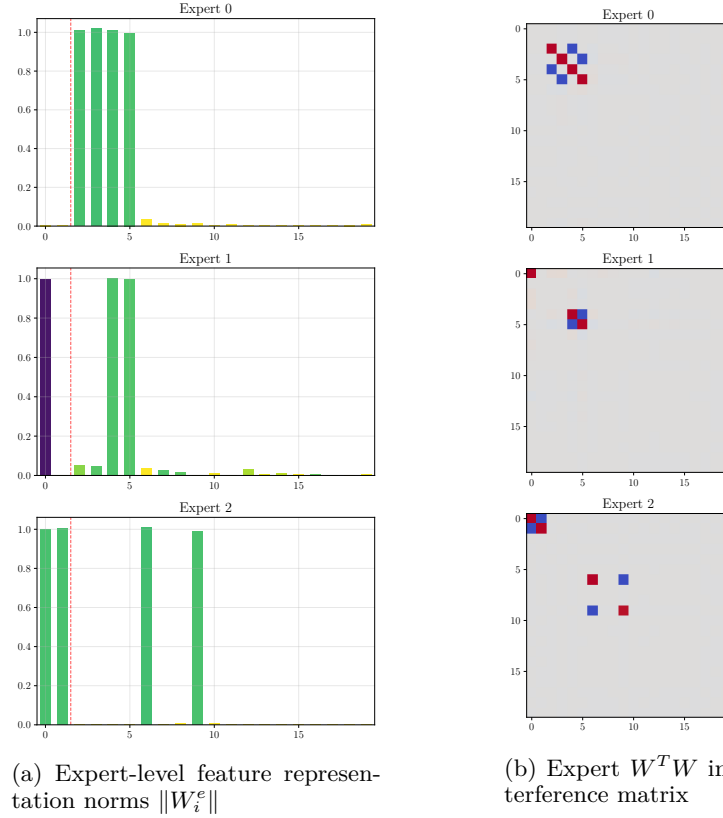


Figure 2: Feature representation and superposition in a MoE with  $n = 20$  features, 3 total experts, and  $m = 2$  hidden dimensions per expert (top- $k = 1$  routing), with importance  $I = 0.7^i$  and feature density  $1 - S = 0.1$ . The expert partition sharply localizes interference compared with the dense model.

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where  $\|W\|_F^2$  is the Frobenius norm and  $p_e$  is the expected probability that expert  $e$  is used across a batch of inputs, i.e. the average renormalized gating weight after top- $k$  routing.

For MoEs with the *same* total parameter count as the dense model, the dense model has more features per dimension (Figure 3) and therefore more superposition. The dense model uses superposition to represent more total features ( $n_{\text{learned}} > m_{\text{total}}$ ), whereas the MoE stays close to the monosemantic limit ( $n_{\text{learned}} \approx m_{\text{total}}$ ). As we increase expert count while keeping total parameters and  $k/E$  roughly fixed, features per dimension decrease. *More experts produce less superposition.* Equivalently, network sparsity makes features more monosemantic. The dense model’s greater superposition gives it consistently lower reconstruction loss than the MoEs in Figure 6 of Appendix A.1, with differences of  $\sim 0.03 - 0.08$  at a given sparsity. However, the gap narrows as expert count grows, and high-expert MoEs reach comparable loss while retaining more interpretable representations. Appendix A.2 gives theoretical intuition for this pattern.

## 5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute those features across experts with markedly less interference. This shows that network spar-

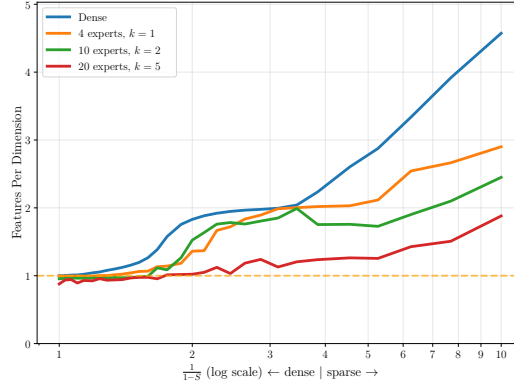


Figure 3: Features per dimension versus inverse feature density ( $\frac{1}{1-g}$ ) for **dense** and **MoE** architectures with uniform feature importance ( $I_i = 1.0$ ). The dense model ( $n = 100$ ,  $m = 20$ ) exhibits the greatest superposition, and superposition decreases decisively as expert count increases: 4 experts with  $m = 5$ ,  $k = 1$  (orange); 10 experts with  $m = 2$ ,  $k = 2$  (green); 20 experts with  $m = 1$ ,  $k = 5$  (red). All models have equal total parameters and similar  $k/E$ . The dashed line at 1.0 marks monosemantic representation.

sity primarily reshapes feature allocation, not merely feature count. We therefore examine whether input-distribution properties—feature sparsity and feature importance—still induce the physics-inspired *phase changes* observed in dense representations.

With finite representational capacity, a model must either ignore a feature, represent it monosemantically, or place it in superposition. A phase change occurs when the boundary between these regimes is discrete as a function of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). Varying feature sparsity and relative importance elicits distinct dense-model strategies; for example, greater feature sparsity encourages greater superposition. Expert-specific MoE phase diagrams reveal a different pattern: experts do not simply reproduce dense-model phase behavior, but adopt routing-dependent representational strategies.

**Setup.** We use the setup from Section 4.1, adding load-balancing loss only where needed to avoid specialization collapse, i.e. local minima in which routers ignore some experts entirely. We study three one-active-expert ( $k = 1$ ) settings: (A)  $n=2$ ,  $m=1$ ; (B)  $n=3$ ,  $m=1$ ; and (C)  $n=3$ ,  $m=2$ . For each setting, we report expert-specific phase diagrams over feature sparsity and last-feature relative importance while increasing network sparsity by raising the number of experts ( $E$ ) up to the number of input feature dimensions ( $n$ ).

Here we fix active parameters rather than total parameters, with  $m_{dense} = km_{moe}$  (ignoring router parameters), so the comparison isolates behavior within each architecture. Otherwise, differences could be attributed to active-capacity changes rather than routing. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. The latter, however, has only one hidden dimension ( $m = 1$ ) for the same number of input features ( $n = 3$ ), whereas the former has two hidden dimensions ( $m = 2$ ), so superposition alone does not explain specialization across these settings.

**Results and Takeaways.** Every single-expert (dense) case shows a clear phase change (Figure 4.X.1/1), reproducing Elhage et al. (2022). Increasing the number of experts removes these discrete phase changes. Some  $E = 2$  MoE experts resemble their corresponding dense cases (Figure 4.X.2/2), but their transitions are visibly smoother. In each architecture, one expert becomes more monosemantic and specializes in the feature with highest relative importance. Experts that diverge from the dense case have uniformly lower superposition scores (bluer regions), confirming the previous section’s central result: MoEs favor lower-superposition representations than comparable dense models.

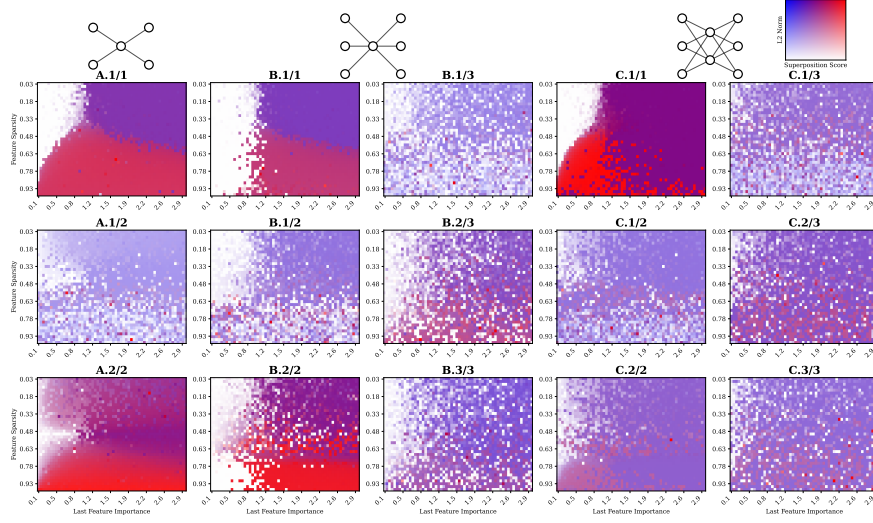


Figure 4: Expert-level phase diagrams decode how a given input dimension (feature) is embedded: ignored (white), monosemantic (blue-purple), or superimposed (red). We plot joint feature norm ( $\|W_n\|^2$ ) and superposition score ( $\sum_{j < n} (\hat{W}_n \cdot W_j)^2$ ) across feature sparsity  $S \in [0.1, 1]$  and relative last-feature importance  $I_n \in [0.1, 3]$ , where  $n$  denotes the last of  $n$  features. For each cell, we train ten models and select the lowest-loss run; this section uses load-balancing loss. Low L2 norm ( $\|W_n\|$ ) appears white and indicates that the model ignores the last feature; otherwise, low superposition appears blue-purple and indicates monosemantic representation, while red indicates superposition. Cell  $(i, j)$  in sub-figure X.e/E denotes expert  $e$  of  $E$  total experts trained on architecture X for last-feature importance  $I_n = i$  and sparsity  $S = j$ ; X.1/1 denotes a dense model.

In the  $n = 2, m = 1$  setting (Figure 4.A), the dense model drops the last feature at low feature sparsity, whereas the comparable MoE preserves it substantially more because routing supplies the needed conditional capacity. With three input dimensions (Figure 4.B), MoEs no longer show this preservation pattern because experts must superimpose the other two features, leaving no room for the third feature in one hidden dimension. Architecture B is therefore the case where one hidden dimension plus superposition is still insufficient, even in the high-sparsity regime, to represent all features. Nevertheless, it still lacks a clean dense-style phase change, except for the mostly discrete but mixed 0.5–0.7 feature-sparsity region in 4.B.2/2. For  $m = 2$ , the dense model’s white region (Figure 4.C.1/1) ignores the last feature in the mid- to low-feature-sparsity regime when that feature has lower relative importance. As network sparsity increases across the remaining Figure 4.C panels, the models represent the last feature with greater L2 magnitude ( $\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$ ). Thus, dimensionality in the low-relative-importance region increases with network sparsity, matching the pattern in Figure 3.

Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 also reveal a feature-sparsity window around 0.48–0.7 where polysemanticity, monosemanticity, and feature ignorance coexist. This mixed region indicates a MoE-specific middle regime in which polysemantic and monosemantic solutions have comparable loss, preventing the model from consistently committing to the strategies seen at lower and higher feature sparsity. Dense models show no analogous pattern, strengthening the conclusion that MoEs learn distinct representational strategies.

**Conclusion.** All experiments in this section use  $\text{top-}k = 1$ , so only one expert is active per input. Even under this dense-like restriction, the  $E = 2$  case already behaves qualitatively differently, including when hidden-dimensional capacity with superposition is sufficient to represent all features. MoEs are therefore not merely aggregations of dense models. The router restricts each expert to a subset of the feature domain, and that restriction decisively changes expert learning and representation.

## 6 EXPERT SPECIALIZATION

Having established that MoEs reduce superposition, we next characterize how these more monosemantic features organize within experts and what that organization implies for specialization.

MoE specialization is often evaluated through load balancing across experts (Chaudhari et al., 2025). Load balancing is useful operationally, but it misses the central semantic criterion: a specialized expert should be used precisely when the concepts it represents are present in the input.

We therefore define specialization by two feature-level conditions: an expert *occupies* particular feature directions in input space, and it represents those features relatively monosemantically. We show that these conditions are directly correlated and that, when they hold, load balancing emerges as a consequence rather than as the definition of specialization.

Because we fix  $k = 1$ , the router partitions feature space into convex cone regions (Appendix A.3), with each region assigned to one expert. By definition,  $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$ , where  $C$  is the set of points inside the cone and  $s$  is any positive scalar. Thus, if feature vector  $\vec{x}$  routes to an expert, every scaled vector  $s\vec{x}$  routes to that same expert. We say that feature  $x$  is contained within expert  $e$ , and therefore expert  $e$  *occupies*  $x$ .

Empirically, small models that distribute input space across more experts achieve lower loss (Appendix A.4). This motivates the central specialization question: does assigning an input-space region to an expert predict that expert’s representational geometry? Our definition predicts that occupancy should align with *monosemanticity*, and we verify this correlation in larger toy models (e.g.,  $m = 10$ ).

For models with  $n > 2$ , we test both directions of this relationship. First, we ask whether initializing experts to occupy specific input-space features makes those features more monosemantic within the corresponding experts. Second, for features that an expert learns monosemantically, we test whether that expert also occupies those features in input space.

With main-diagonal gate initialization, each expert monosemantically represents exactly the single feature it initially occupied (Figure 5a). With ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b). The remaining experts, initialized over less important features, do not monosemantically represent those features, and they also do not acquire the five most important features they were not initialized to occupy. When feature-importance ordering is broken and each expert initially occupies random features, each expert monosemantically represents only the most important feature(s) within its initialized set (Figure 5c).

These results show a strong alignment between initial routing occupancy and eventual monosemantic representation. They also show that importance controls which occupied features become monosemantic: experts devote clean dimensions to important features, whether each expert is initialized with one important feature or with a set from which it selects the most important feature(s).

Under uniform feature importance, experts place all features in superposition with higher interference. Even in this harder setting, initial occupancy still matters: features initially occupied by an expert are *relatively* more monosemantic, with superposition scores 1.03 standard deviations below that expert’s average.

We then test the reverse direction: whether experts that represent features monosemantically also occupy those features in input space. We measure expert usage when the corresponding feature(s) are *one of many* active features and when they are the *only* active features; the second condition directly estimates occupancy probability. Table 1 shows that the correlation holds for both xavier and k-hot initialization. With  $E = 10$ , mean usage near  $\sim 10\%$  indicates balanced average load. Conditional on an expert’s monosemantic feature(s) being active, expert usage rises sharply; when those feature(s) are the only active features, the expert dominates routing. Under k-hot initialization, *100%* of features represented monosemantically by an expert are occupied by that same expert.



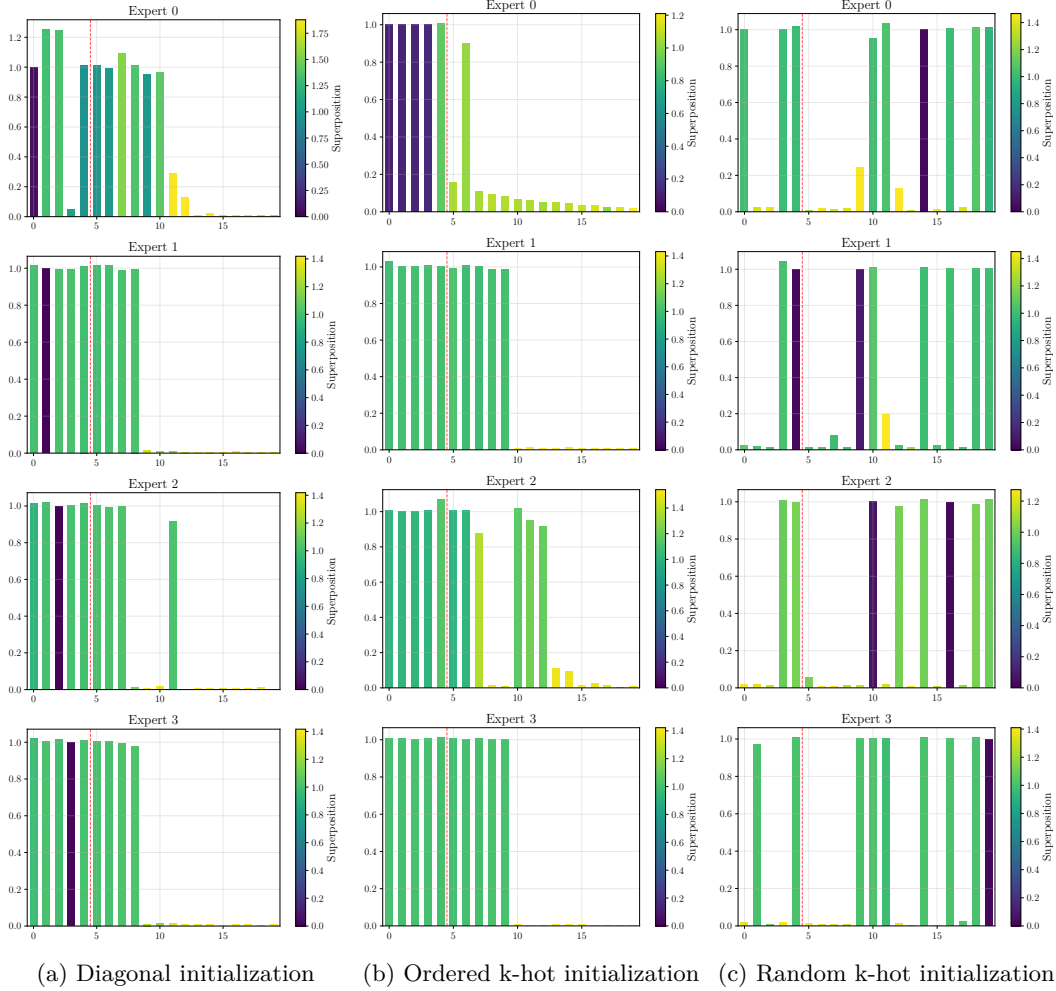


Figure 5: Expert feature norms  $\|W_i^{(e)}\|$  and superposition (color) under three initialization schemes, with  $n = 20, m = 5, E = 4, S = 0.1$ . In **(a)**, the gate matrix is initialized along the main diagonal ( $W_i^r = \hat{e}_i$ , the basis vector for that dimension), and relative feature importance decreases exponentially from feature one to feature 20. In **(b)**, the gate matrix uses “ordered k-hot” initialization: the first expert aligns with the first five features, and each subsequent expert aligns with the next five features. Relative feature importance matches **(a)**. In **(c)**, the gate matrix uses “random k-hot” initialization, assigning each expert five random features with no feature overlap across experts and full coverage of all 20 features. Relative feature importance still decreases exponentially but is randomly assigned across features.

Experts that represent more features monosemantically are therefore more specialized in the feature-based sense. Their average usage on arbitrary inputs can decrease, but their conditional usage rises far above baseline when their specialized features are active. This pattern holds across the table except for the xavier-initialized model with one four-feature monosemantic expert, where utilization drops substantially.

## 7 CONCLUSION

We show that experts change superposition in a systematic and interpretable way: MoEs consistently exhibit greater monosemanticity than comparable dense networks and replace dense-model phase changes with smoother, routing-mediated transitions. We also introduce

Table 1: Monosemantic feature and usage statistics per expert for  $n = 100, m = 10, E = 10$ . We train one hundred models for each initialization scheme (xavier and k-hot), yielding 1000 experts per scheme. Statistics are aggregated across models after grouping experts by the number of features they represent monosemantically. For each expert’s monosemantic feature(s), we report usage when those feature(s) are one of several active features and when those feature(s) are the *only* active feature(s), directly measuring the link between monosemantic representation and routing occupancy.

| Xavier Initialization                      |                                 |                       |                                          |                                               |
|--------------------------------------------|---------------------------------|-----------------------|------------------------------------------|-----------------------------------------------|
| Number of monosemantic features per expert | Number of experts (out of 1000) | Mean expert usage (%) | Mean expert usage; feature(s) active (%) | Mean expert usage; only feature(s) active (%) |
| 0                                          | 461                             | –                     | –                                        | –                                             |
| 1                                          | 387                             | 9.595                 | 17.94                                    | 67.18                                         |
| 2                                          | 138                             | 9.599                 | 30.29                                    | 95.65                                         |
| 3                                          | 13                              | 8.363                 | 40.19                                    | 100.0                                         |
| 4                                          | 1                               | 1.428                 | 14.69                                    | 100.0                                         |
| 5                                          | 0                               | –                     | –                                        | –                                             |
| K-Hot Initialization                       |                                 |                       |                                          |                                               |
| Number of monosemantic features per expert | Number of experts (out of 1000) | Mean expert usage (%) | Mean expert usage feature(s) active (%)  | Mean expert usage only feature(s) active (%)  |
| 0                                          | 335                             | –                     | –                                        | –                                             |
| 1                                          | 382                             | 10.00                 | 23.94                                    | 100.0                                         |
| 2                                          | 227                             | 10.02                 | 46.61                                    | 100.0                                         |
| 3                                          | 47                              | 10.09                 | 62.00                                    | 100.0                                         |
| 4                                          | 8                               | 9.95                  | 70.30                                    | 100.0                                         |
| 5                                          | 1                               | 9.62                  | 74.79                                    | 100.0                                         |

a feature-based definition of expert specialization and demonstrate that experts organize around coherent features when initialization supports that organization. These claims are grounded in simple autoencoder toy models with synthetic data, so generalization to large-scale transformers with unknown feature distributions remains an open question (see Appendix A.6). Within this controlled setting, however, the result is clear: toy MoEs achieve comparable reconstruction loss while maintaining more interpretable representations. This directly challenges the prevailing view that mechanistic interpretability and model capability are fundamentally in tension. Future work should identify which MoE design choices most reliably favor monosemanticity, how MoE training dynamics diverge from dense-model dynamics, and when specialization emerges at scale. Answering these questions can guide the design of language models that are both high-performing and more interpretable.

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## A APPENDIX

### A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

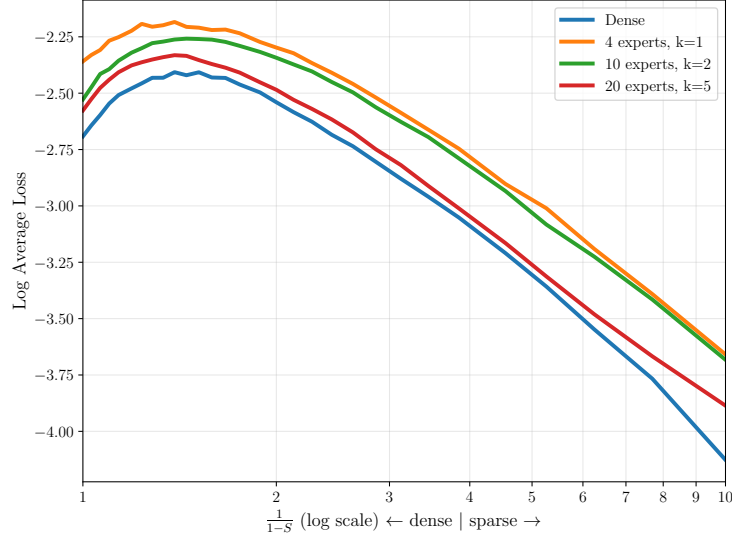


Figure 6: Log average loss versus feature density ( $\frac{1}{1-S}$ ) for dense ( $m = 20$ ) and MoE (4 experts,  $k = 1$ ,  $m = 5$ ), MoE (10 experts,  $k = 2$ ,  $m = 2$ ), and MoE (20 experts,  $k = 5$ ,  $m = 1$ ) models, all with uniform feature importance ( $I_i = 1.0$ ) for  $n = 100$  input features. Results are averaged over five runs per sparsity level. The dense model has lower loss than all MoEs at every sparsity level, but the gap consistently narrows as expert count increases.

### A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition exploits the gap between sparse features and dense computation by compressing rarely active features into shared dimensions. MoEs address the same gap structurally through conditional computation. By aligning activation sparsity with feature sparsity, MoEs reduce the computational penalty of dedicating capacity to rarely active features. As Elhage et al. (2022) note, when a model only expends computation on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes optimal; MoEs operationalize this trade by replacing the *compression* of superposition with the *selection* of routing.

Geometrically, this structural change appears as reduced *interference*. Even when the global ratio of represented features to parameters remains fixed, the router partitions feature space so that a feature routed to expert  $e$  competes only with the subset of features assigned to that expert. Interference is therefore governed by the local matrix  $(W^e)^\top W^e$  rather than the dense model’s global  $W^\top W$ . This partitioning sharply reduces the number of interfering features for any feature vector, weakens the optimization pressure toward superposition, and enables experts to learn more monosemantic representations.

### A.3 ROUTER SUBSPACES ARE CONVEX CONES

When  $k = 1$ , the router  $g(x) = \text{softmax}(W^r x)$  selects the same expert as  $\text{argmax}(W^r x)$ . The region routed to expert  $i$  is  $\forall j \neq i, (w_i - w_j)^\top x > 0$ , where  $w_i$  and  $w_j$  are rows of  $W^r$ . This homogeneous linear inequality defines a convex cone. If  $x$  satisfies it, then multiplying by any positive scalar  $s$  also satisfies it; if  $x_1$  and  $x_2$  satisfy it, then any convex combination  $x = x_1 \lambda + (1 - \lambda)x_2$  for  $\lambda \in [0, 1]$  also satisfies it.

For  $k > 1$ , the inputs sent to a particular expert  $e$  form a union of convex cones. A union of convex cones is not generally itself a convex cone, so the clean occupancy interpretation we use for  $k = 1$  may not extend directly to multi-expert routing.

#### A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

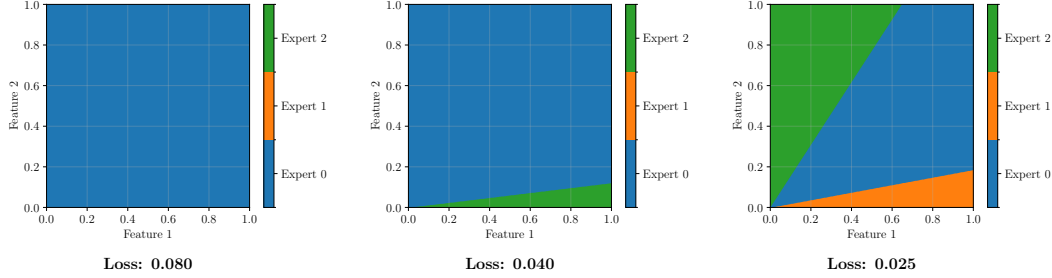


Figure 7: Expert routing for three identical models with different initialization schemes. We use  $n = 2$ ,  $E = 3$ ,  $m = 1$ . The first model (left) performs worst (loss: 0.08) and routes all inputs to one expert. The second model (middle) performs better (loss: 0.04) and routes a small input region—cases where feature 1 is active—to a second expert. The third model (right) performs best (loss: 0.025) and distributes the input space across all experts: one expert handles feature 1 alone, one handles feature 2 alone, and one handles both active features.

In small models ( $n = 2, m = 1, E > 1$ ), models that distribute input space across more experts consistently tend to achieve lower loss across  $E \in [2, 7]$ . Fixing  $n = 2$  makes the routing geometry directly visible, as shown in Figure 7.

#### A.5 ANALYTIC MODEL EQUIVARIANCE

For the single-layer, single-nonlinearity, top- $k = 1$  toy MoE, a theoretical map exists between any dense model and a monosemantic MoE with an equivalent number of active features under a sparsity constraint.

Assume an upper bound  $a$  on active features for every input, so  $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$ . Also assume  $a$  is no greater than expert hidden dimensionality  $m$ , giving each expert enough dimensions for the active features it must represent, and assume  $m$  is smaller than the total number of input features  $n$  ( $a \leq m \leq n$ ). The construction is direct: for each subset  $S \subseteq \{1, 2, \dots, n\}$  with  $|S| \leq a$ —that is, each possible active-feature subset of size at most  $a$ —create an expert that preserves those features monosemantically. (It is sufficient to use only subsets with  $|S| = a$ .) The router then selects the expert corresponding to the active features, which by assumption never exceed  $a$ :

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where  $\text{support}(x) = \{i : x_i \neq 0\}$ . Since  $|S| \leq a \leq m$ , every expert has enough capacity to represent its assigned features without superposition. Because at most  $a$  features are active, each unique active-feature combination receives a dedicated expert with sufficient monosemantic capacity. The number of possible experts needed is therefore  $\binom{n}{a}$ .

This theoretical MoE achieves zero reconstruction loss when the sparsity constraint holds (or approaches one in these toy models). If more than  $m$  features can be active at once ( $a \not\leq m$ ), the input exceeds the monosemantic capacity of a single expert; absent favorable feature correlations, a dense polysemantic model faces the same capacity problem. Thus, even when  $a \not\leq m$  occasionally, the constructed monosemantic MoE can still remain competitive or outperform under looser sparsity constraints.

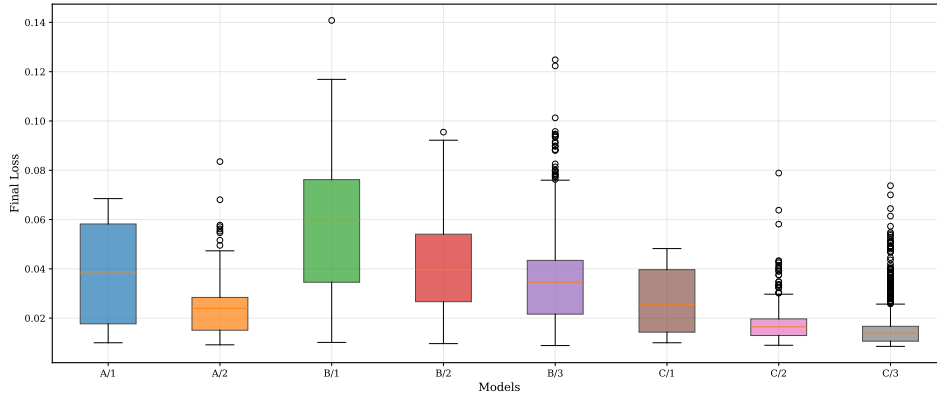


Figure 8: Model  $X/E$  uses  $X$  for the architectures from Figure 4 and  $E$  for the total number of experts (i.e. network sparsity). Increasing network sparsity lowers mean loss while increasing localized variance, especially when the number of experts reaches the number of input feature dimensions. This pattern reflects the relatively unstable training of MoEs compared with dense models, even though we train ten models per cell and select the lowest-loss run.

Therefore, for any dense model  $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$  under the sparsity constraint  $|\text{support}(x)| \leq a$ , there exists an MoE model  $f_{\text{MoE}}(x)$  such that  $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$  for all valid inputs. In our toy settings, this sparsity constraint holds in the limit as sparsity goes to one. In practice, meaningful data may impose an upper bound on how many features a fixed amount of information can semantically encode, reflected in the size of useful embeddings for that data. A MoE with enough experts and a tractable amount of superposition may therefore be sufficient to encode all present features while remaining interpretable.

## A.6 LIMITATIONS

Our claims are intentionally scoped by several limitations. First, every experiment uses controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). This design enables precise measurement of interference and monosemanticity, but it omits many large-transformer complexities, including multi-step computation, attention, heterogeneous feature distributions, and realistic routing dynamics. The transfer of these patterns to real-world MoE architectures therefore remains an empirical question.

Second, our architectural and routing choices are deliberately restricted: simple experts, fixed top- $k$  routing (often  $k = 1$ ), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses except where stated. Practical MoEs often use richer routers, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), and they typically optimize objectives such as next-token prediction rather than reconstruction alone. These differences may produce qualitatively different representational behavior.

Third, our synthetic feature distributions are far simpler than natural data: sparsity patterns, feature importance, and feature independence are all controlled rather than learned from realistic corpora.

Finally, although we robustly observe increased monosemanticity and reduced superposition in these fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work indicates that these dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, the study provides a clean mechanistic account of how network sparsity reduces superposition in toy MoEs, while leaving large-scale validation as the essential next step.

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810 Our current measurement methods do not scale naturally to larger models, making scalable  
811 diagnostics for MoE monosemanticity a clear priority for future work.  
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